

Review

A Literature Review on Inventory Pooling with Applications

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Abstract: In this paper, we provide a review of academic research on inventory pooling published between 2010 and 2024, with a particular emphasis on studies that focus on real-world applications. The review analyzes the research conducted over the past 14 years, evaluates the outcomes of these applied studies, and identifies gaps in the literature. The contribution of this work is twofold: firstly, it provides insights into the extent to which theoretical advancements in inventory pooling have been implemented in the practice; secondly, it provides practitioners with an overview of recent real-world applications across various industrial contexts. The findings highlight the impact of inventory pooling on cost savings, service level improvements, inventory optimization in diverse sectors, and sustainability. Additionally, this paper examines the contributions of inventory pooling to economic, environmental, and social sustainability, offering a comprehensive analysis of its role in fostering sustainable practices across supply chains. Finally, the paper discusses practical challenges encountered in implementation and suggests directions for future research in this domain.

Keywords: inventory pooling; inventory sharing; applications in practice; sustainability; sustainable practices

1. Introduction

Inventory pooling is an important strategy recognized for its potential to provide cost savings while optimizing the inventory levels in a coordinated manner in supply chains. At its core, inventory pooling is a practice of consolidating inventory from multiple locations or products into a shared stockpile, which can be used by all participating actors. The main purpose of this approach is to leverage the relatively large size of the pool to buffer against demand variability and, therefore, minimize the risk of shortages and reduce the overall safety stock required. Depending on the concept, the inventory pool can be managed by a single decision maker or managed by a coalition formed by different entities and firms. In both cases, by pooling inventory, organizations aim to achieve higher service levels and lower holding costs, so that significant operational efficiencies can be obtained.

Inventory pooling is not only a strategic tool for cost reduction and operational efficiency but also has the potential to contribute to sustainability within supply chains. By consolidating inventory from multiple locations or products, inventory pooling can help reduce the need for excess stock and, as a result, lower overall resource consumption. This reduction in stock levels may lead to fewer storage and transportation requirements, which could help decrease carbon emissions and environmental impact. Additionally, centralizing inventory management can optimize the use of available resources, potentially reducing waste from product obsolescence and overproduction. In this way, inventory pooling aligns with sustainable practices by improving supply chain efficiency, potentially reducing environmental footprints, and contributing to long-term resource conservation.



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Through these measures, businesses could achieve a balance between profitability and environmental responsibility, supporting both economic and sustainability goals.

The concept of inventory pooling has been extensively studied in the literature since the pioneering work of Eppen [1], and it continues to attract significant attention in both academia and industry. There is a substantial amount of work on reviews in the concept of inventory pooling, risk pooling, and collaboration in stock management, among others. We refer the reader to the relevant past reviews [2–8]. These studies cover a diverse range of pooling-related topics, including component commonality and its effects on costs, risk pooling in supply chains, commonality in manufacturing resources, inventory models with lateral transshipments, optimal pooling of inventories with substitution, supply chain pooling, and closed-loop repairable parts inventory systems, respectively.

Recent literature reviews that encompass related concepts of inventory pooling include [9–13]. However, these reviews primarily focus on specific dimensions of inventory management or emphasize particular research methodologies. For instance, refs. [9,11,13] review the literature on spare parts inventory management. In [9], inventory centralization in spare parts management is addressed, while [13], with a predominant focus on recent technological advancements, review spare parts management in Industry 4.0 era. In [10], research-driven behavioral experiments related to inventory and ordering decisions are examined. Finally, ref. [12] explores coordination, cooperation, and collaboration with a particular emphasis on production-inventory systems. Therefore, the majority of these reviews offer only partial and limited insights into the real-world applications of inventory pooling studies.

Existing theoretical models and the reviews in the literature highlighted the potential benefits of inventory pooling. However, the true value of this methodology can only be fully appreciated when it is applied in practice. The proper and successful implementation of inventory pooling in real-life situations requires a thorough understanding of the specific context, including product types, supply chain dynamics, and demand characteristics. Consequently, real-life applications offer empirical evidence on how these strategies can be effectively adapted to address the challenges experienced by different industries. Moreover, investigating the real applications of inventory pooling is important to bridge the gap between theory and practice.

Our aim in this review is to provide an overview of inventory pooling with a particular emphasis on recent trends and the real-world applications in the field, as well as its integration with sustainable practices. To this end, we have reviewed the studies that consider real-world applications and were published in the past 14 years. By analyzing the findings of these studies, we aim to demonstrate how theoretical advancements in inventory pooling have been applied in practice and to provide valuable insights for professionals. This review contributes to the literature in two different ways: firstly, by highlighting the intersection of theory and practice in inventory pooling, and secondly, by offering a practical guide for professionals planning to implement efficient inventory pooling strategies within their organizations. Furthermore, this review aims to guide practitioners in increasing the adoption of inventory pooling applications, ensuring that these strategies not only enhance operational efficiency but also contribute to more sustainable practices by reducing waste, optimizing resource utilization, and lowering environmental impacts.

The outline of the paper is as follows: in Section 2, we describe our review methodology, including the tools used to retrieve the relevant research and the techniques used to refine our search. Section 3 presents an analysis and classification of the reviewed studies. We identify the gaps in the literature and provide potential areas for future research in Section 4. Finally, Section 5 concludes the paper by summarizing our findings.

2. Methodology

In order to review the studies published between 2010 and 2024, we used the Web of Science platform. We used the keyword “inventory pooling” and restricted the year field to 2010–2024. The initial search resulted in a total of 2798 papers. Then, we made further refinements on the research field and included “Operations Research Management Science”, “Engineering”, “Business Economics”, “Science Technology Other Topics”, “Computer Science”, “Social Sciences Other Topics”, “Mathematics”, and “Materials Science” to cover a wide range of applications in the field. Then, we obtained a total of 646 research items. As a final refinement, the conference papers, proceedings, and Web of Science “Emerging Sources Citation” indexed papers are excluded. As a result, we obtained a total of 520 research items in the list.

In our work, in order to find the studies that consider the real-world applications, we examined 520 studies listed after the initial refinement. We selected those that included real-world applications or used real-world data as part of the research. Elimination in this step requires a lot of effort since each paper is examined in a detailed way to seek if they applied proposed models to the real-life settings or used real-life data in their research. Papers that use real-life data from existing articles to create benchmark studies are also included. The eligibility criteria for the articles chosen, therefore, consist of the real implementations in practice, real-life data usage for potential implementations, and the case studies using real-life data. After evaluation, we obtained a total of 54 papers, and further analysis and classifications are provided in Section 3. Figure 1 depicts the PRISMA flow diagram for the article selection process in this review. Figure 2 shows the article count based on the publication years.

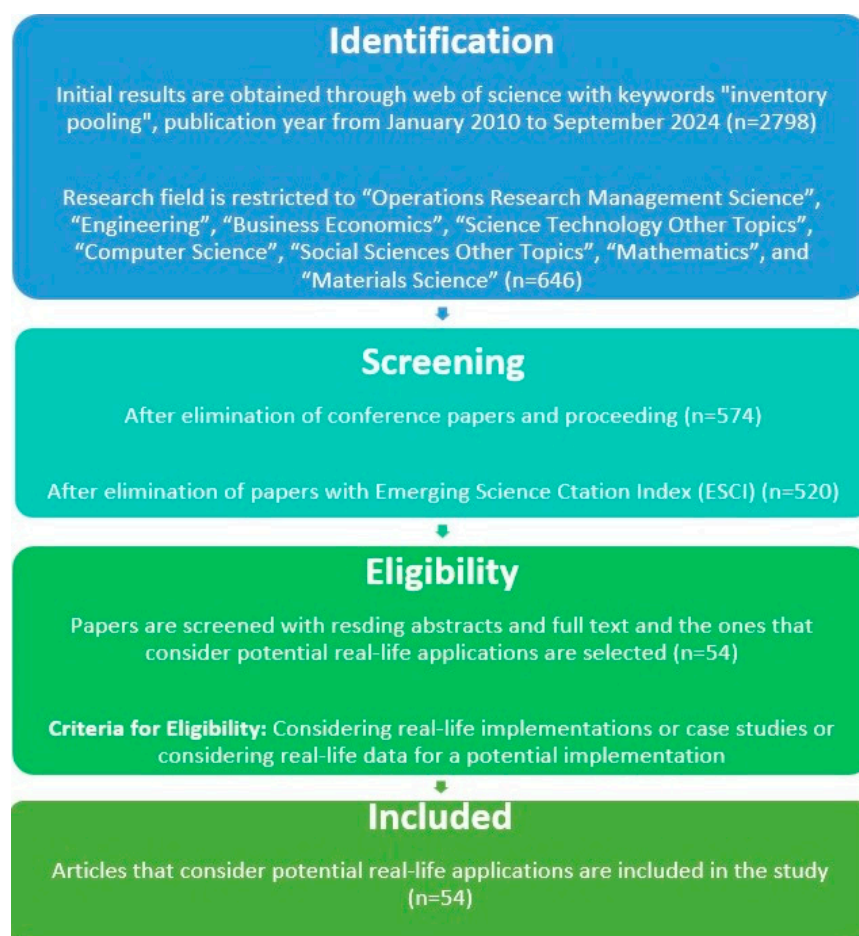


Figure 1. PRISMA flow diagram for literature search and selection process.

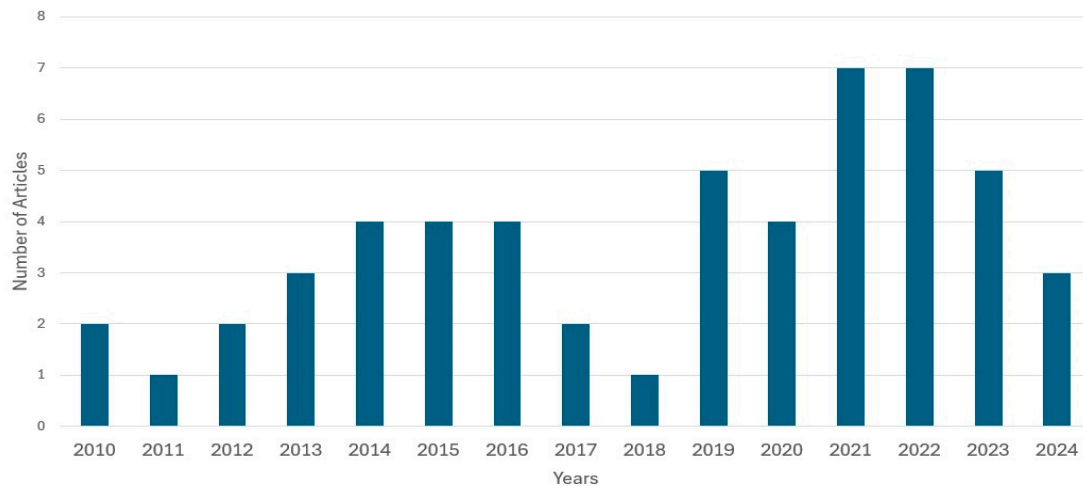


Figure 2. Article numbers based on publication years.

3. Analysis and Classification

In this review, we initially categorize the related work into two distinct groups: the first one consists of empirical research studies, and the second one encompasses original research articles. In this classification, empirical research papers refer to the studies that conduct behavioral experiments, surveys, interviews, etc., and then qualitatively or quantitatively analyze the findings. These articles aim to investigate whether the real-life collected data or experience supports some specific hypotheses. Therefore, these works also shed light on how theory can be adapted into the practical world. Figure 3 depicts a pie chart of this classification, and the majority of the articles belong to the original research category. In total, 9 out of 54 papers are classified as empirical research studies. The remaining 45 articles are described as original research studies, and besides including potential real-life applications, they provide new models or solution techniques.

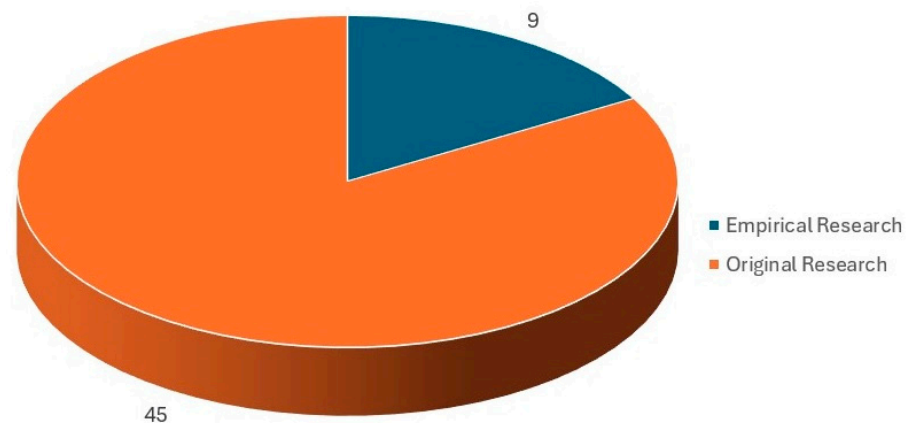


Figure 3. Article numbers based on research type.

3.1. Sector-Based Applications

In this review, we consider the articles that include real-world applications in the concept of inventory pooling. Therefore, we provide a categorization of the reviewed articles based on the sector. This classification offers practitioners an insight into which sectors have potential real-life implementations of the inventory pooling concept. We categorize the sector of empirical research papers as various since depending on the used research methodology, the sector varies significantly. Figure 4 shows the sector-based distribution of articles, and healthcare is the leading sector.

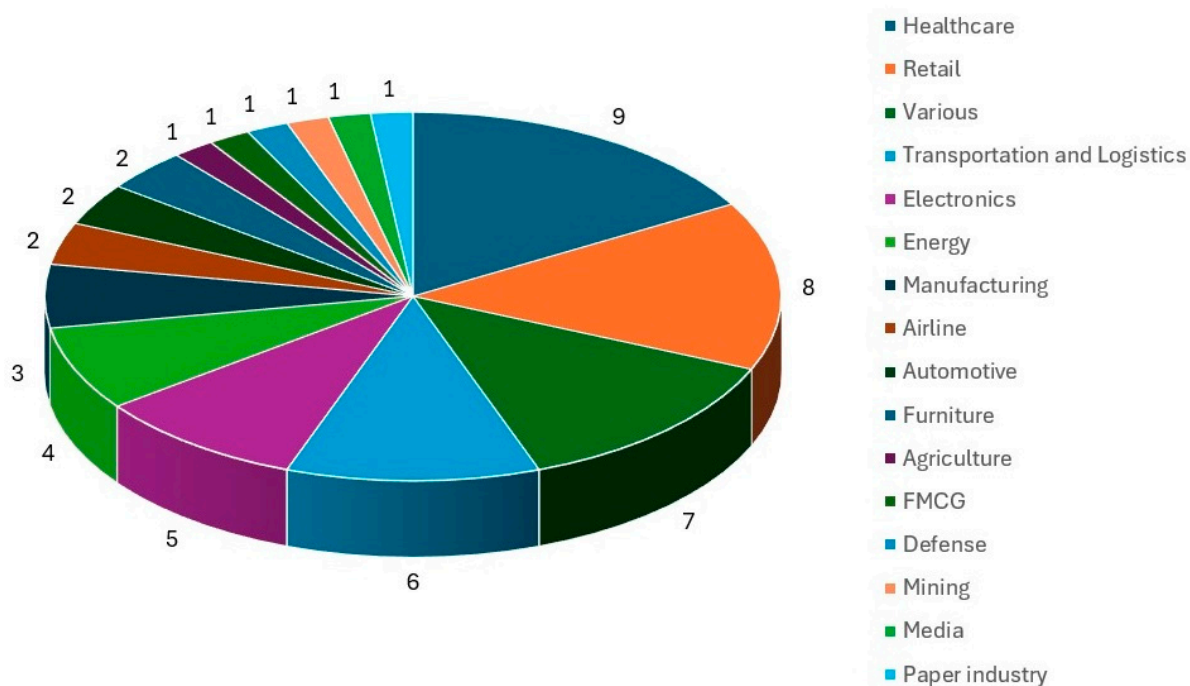


Figure 4. Article numbers based on different sectors.

In the classification, apart from various sectors, which are associated with empirical research studies, three leading sectors emerge: healthcare, retail and transportation, and logistics. Studies that fall in the healthcare category generally consider blood or pharmaceutical products for inventory pooling. While the retail sector group encompasses a diverse range of products, in the transportation and logistics sector category, papers that focus on pallet industry products stand out. The healthcare sector is particularly well-suited for inventory pooling due to the high demand variability and perishable nature of certain products, such as blood or pharmaceuticals, which require precise management to ensure availability while minimizing waste. Similarly, in retail, demand fluctuations across different regions and the need for efficient logistics in managing stock across multiple stores make inventory pooling a valuable solution to optimize inventory levels and reduce costs.

3.2. Product/Service Type-Based Applications

Inventory pooling holds significant potential for implementation in a wide range of practical problems across different organizations. The bar chart in Figure 5 categorizes various product types considered in potential real-world applications of inventory pooling based on the number of articles reviewed. The leading category labeled as “Diverse” in this chart comes forward with 18 articles. The majority of these include empirical research studies and the ones that consider problems in the concept of retailing. In this category, the product types are not specific, and empirical studies generally include a wide range of industry surveys. Following diverse, spare parts come as a second leading category, and the product types include electronic motors, data storage systems, semiconductor industry, oil and gas industry products, airline industry products, paper industry products, etc. Observing spare parts in different industries emphasizes the critical potential of pooling to ensure availability while reducing the downtime for main products. Drugs come in third place for real-life implementation of inventory pooling. Blood and pallets emerge as significant areas of focus and have notable consideration, too. The other types include reusable transport items, batteries, and furniture. The remaining articles consider different ranges of products, from pet food and care products to fashion goods.

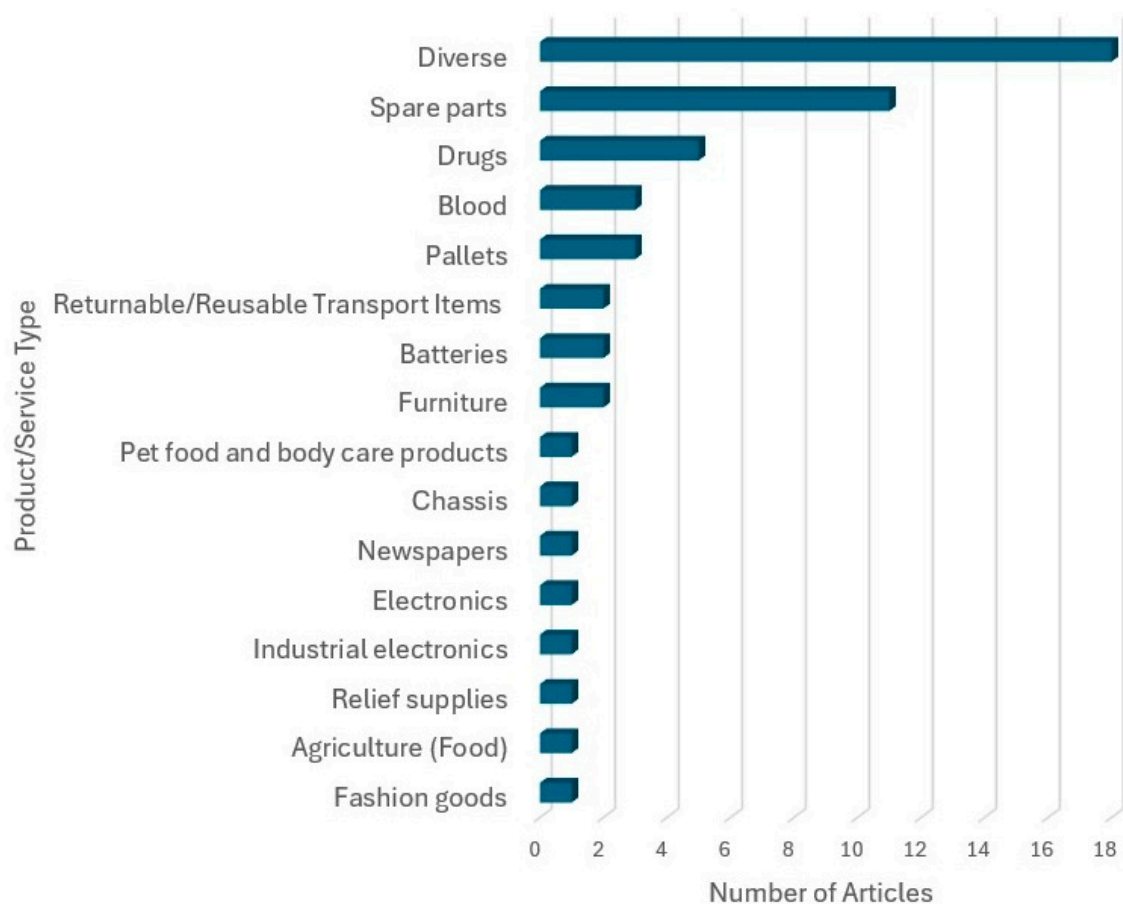


Figure 5. Article numbers based on different product or service types considered.

The distribution of the product types across studies shows that inventory pooling is considered across a wide range of industries. However, certain product types that are critical, perishable, or involved in chains with logistical complexity have prominent potential in possible implementations of inventory pooling strategy. Organizations managing the products with these specific characteristics may consider implementing pooling within their operations to explore its potential benefits.

3.3. Data Descriptions Used in the Articles

In this section, we present the detailed descriptions of the data used in all articles in Table 1. We can see that among others, hospitals, retailers, and the companies that provide transportation and logistics services stand out prominently. In this review, we consider both studies that present real-life applications and the ones utilized real-life data and create realistic case studies. In this way, we aim to present a more comprehensive overview of related work published in the past 14 years and, therefore, provide practitioners with valuable insights for potential applications as well.

Table 1. Data descriptions that are used in the articles.

Work	Data Description
[14]	Data from United States Air Force bases
[15]	Data from Applied Materials, a large supplier for the semiconductor industry in North America
[16]	Data from 38 Italian airports

Table 1. Cont.

Work	Data Description
[17]	Model is applied for a potential retail chain that plans to enter the retail market in Toronto, Ontario
[18]	Model is considered for a potential application using a 30-city dataset from other existing studies
[19]	A spare parts pooling project involving five different paper-making industries in Italy
[20]	A clinical trial data (drug safety stocks) from other existing studies
[21]	Data from 16 United Kingdom hospitals (transfusion laboratories)
[22]	Data from the industrial electronics industry
[23]	Data from copper mining company
[24]	A real-life test case for electric motors
[25]	Data from Yedioth Group sales
[26]	United States Mergers and Acquisition database
[27]	China's public hospitals data supported with simulation
[28]	Data from the oil and gas industry supported with large computational studies
[29]	Data obtained from Statoil ASA, a large energy company
[30]	United States Census data
[31]	European airline industry case study
[32]	Behavioral experimentation conducted at a laboratory
[33]	Data from the pallet industry
[34]	Semi-structured interviews using open-ended interview protocol in French by two researchers
[35]	Data from United States container port
[36]	Battery manufacturing supply chain from other existing studies
[37]	Parameters are taken from literature, and real data obtained from several hospitals
[38]	The United States Census 88-node dataset is used
[39]	Data taken from company visits in West Bengal, India
[40]	Data from partners and public databases (Emergency Events Database, the National Oceanic and Atmospheric Administration database, and Caribbean Hurricane Network)
[41]	Experiments on the 14-stage supply network from other existing study
[42]	Blood distribution network in Istanbul
[43]	Data from a world-leading manufacturer in the semiconductor industry
[44]	Data from reusable transport item pooling company in China
[45]	Data from a real-life data storage system
[46]	Data collected from an online retailer in China
[47]	Unofficial data of a United States-based online retailer's fulfillment center network from other existing studies
[48]	Conducted experiments in the laboratory
[49]	A survey with German Hospitals
[50]	Data from capital equipment manufacturers in North America
[51]	Data from hospital pharmacies in Chile
[52]	Modified data from other existing works that consider real-life data of a large global FMCG
[53]	Conducted experiments

Table 1. *Cont.*

Work	Data Description
[54]	Real-life data about four chemotherapy drugs obtained from Harris Health System in Houston, TX
[55]	Data from drop shipping furniture company
[56]	Realistic data from a shipyard
[57]	Data from an anonymous retailer in North America
[58]	Real-world case of the Indian agricultural system
[59]	Case study for Returnable Transport Items of a company in automotive industry
[60]	Pharmaceutical distribution in Morocco
[61]	Drop-ship manufacturer furniture and home interior
[62]	Omnichannel fashion retailer in Manhattan
[63]	Real data from a gas station in Beijing
[64]	Real demand data from an industrial facility and a residence
[65]	Pallet industry in Heilongjiang Province
[66]	Surveys and interviews in different positions, CEO, distributors, industrialists
[67]	Data from TE-Logistics

The datasets analyzed in the reviewed studies can be categorized into four groups: real-life data, data from existing studies, experimental data, and database-derived data. Among these, the majority of the articles (38 in total) utilized real-life data in their research [14–17,19,21–25,27–29,31,33,35,37,42–46,49–51,54–65,67]. The use of real-life data is particularly important for capturing the dynamic and uncertain nature of real-world operations, such as seasonal fluctuations, sudden demand surges, and regional variations.

Six of the reviewed articles employed data from existing studies [18,20,36,41,47,52]. These datasets, repurposed from prior research, offer structured and curated information, ensuring consistency and accessibility. However, they may lack the variability and unpredictability characteristic of real-world scenarios, potentially limiting their applicability to dynamic environments.

Experimental data, used in six studies [32,34,39,48,53,66], is typically generated in controlled laboratory settings and is valuable for testing theoretical frameworks. Despite its utility in providing proof-of-concept evaluations, such data often fails to account for the complexities and stochastic nature of real-world demand, thereby reducing its relevance in practical applications.

Finally, database-derived data, utilized in four studies [26,30,38,40], offers broad and historical insights, which are useful for identifying long-term trends or analyzing extreme events. However, the static and aggregated nature of such datasets often limits their ability to capture the real-time dynamics and variability of demand, reducing their robustness for operational decision-making.

In summary, while real-life data are particularly valuable for capturing the dynamic and uncertain nature of demand, data from existing studies, experimental data, and database-derived data have certain limitations in fully representing real-world complexities.

3.4. Modeling and Solution Approaches

We analyze the main characteristics of the modeling and solution approaches that are employed in the reviewed articles in this section. Figure 6 presents the distribution of the studies based on the predominant modeling approach they utilized, categorized as stochastic, deterministic, or not applicable (NA).

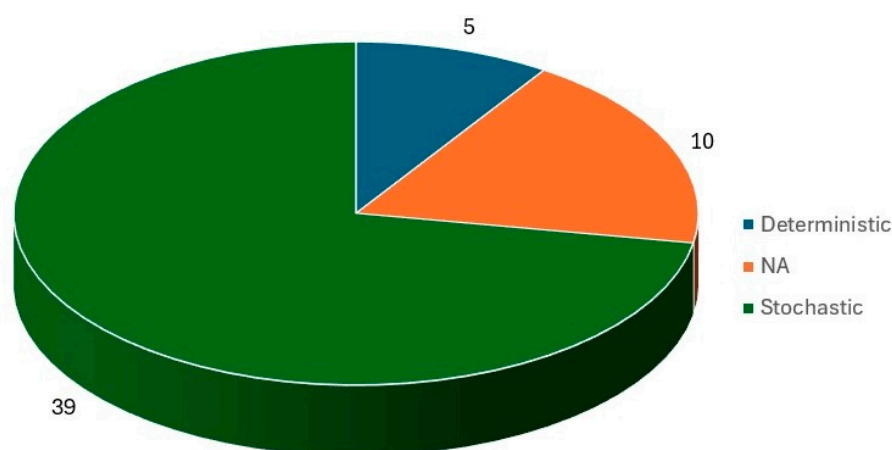


Figure 6. Modeling characteristics (stochastic/deterministic) of the articles reviewed.

Most of the articles choose to implement stochastic models, which aligns with the complex nature of real-life problems involving inventory pooling. The studies that conduct empirical research are classified as not applicable since they rely on behavioral experiments, surveys, and interviews and analyze their results. Relatively few papers, 5 out of 54, used deterministic modeling approaches. In this classification, we focus on whether the models predominantly contain stochastic or deterministic elements when categorizing the reviewed articles. Deterministic models are typically suitable and preferred in scenarios where system parameters are stable, well-known, and not subject to significant uncertainty. Stochastic models, on the other hand, are favored in situations characterized by high levels of uncertainty and dynamic behavior, such as fluctuating demand, variable lead times, or supply chain disruptions. Their flexibility and ability to capture real-world complexities make them particularly valuable for addressing the changing dynamics of inventory pooling.

We provide detailed modeling and solution approach descriptions in Table 2. The distribution of modeling approaches and the solution methods in these studies also vary depending on the inventory pooling context. Articles that are categorized under empirical research studies use qualitative or quantitative methods to obtain their results. While original research articles adapt various modeling techniques with a prominent focus on stochasticity. This is expected as the inventory pooling problems in real life are subject to changing dynamics of supply chains, decision makers, and many other entities. Models with predominant stochasticity are often challenging to solve due to the increased computational effort required to handle randomness and variability. As a result, approximation techniques, simulation models, and heuristic methods are commonly employed as solution approaches for such models, balancing computational feasibility and practical applicability.

A significant portion of the solution approaches used in the original research articles rely on heuristics and approximations. These methods are typically employed to find good approximate solutions when solving problems to optimality is computationally expensive or impractical. This also aligns with the inherent complexity of modeling real-world inventory pooling problems. Considering the modelling approaches in original research articles, we can say that the models involving stochasticity dominate the deterministic ones. Apart from this, there is no distinct tendency towards a particular modelling technique. Mixed integer programming models, analytical models, game theoretical approaches, newsvendor models, and general optimization models are listed among many others.

Table 2. Modeling and solution approach details of reviewed articles.

Work	Modeling Approach	Solution Method
[14]	Inventory allocation and warehouse location (modeled as a p-median problem)	Ardalan's heuristics
[15]	A mixed integer linear programming model	Using commercial solvers and a tool developed by Solvoyo
[16]	Markov Chain Model	Approximations
[17]	Coordinated location–inventory model. A nonlinear integer programming problem	A Lagrangian relaxation algorithm
[18]	Stochastic programming model	Algorithms with Lagrangian relaxation and variable splitting
[19]	Hybrid pooling system with coopetition	Agreement among the 5 companies for fair allocation and the monitoring is supported by using several software tools
[20]	Simulation model	Simulation-based optimization
[21]	Case study methodology with several research questions	Case analysis and surveys
[22]	An exact mathematical model	Adjusted DP algorithm
[23]	A decision-making framework	Combination of optimization and analytical derivations
[24]	A mixed integer nonlinear programming problem (MINLP)	A spatial decomposition algorithm
[25]	A two-stage stochastic optimization	A stochastic gradient-based optimization algorithm
[26]	Empirical research and hypothesis testing on real data	Empirically testing the hypotheses of theoretical models in the mergers and acquisitions context
[27]	Simulation studies	Simulation models solved by using Vensim software
[28]	Game theoretical approach	An allocation method based on the solution to a linear programming model
[29]	Service-level constrained model	Approximation
[30]	Joint location–inventory model, formulating a compact nonlinear mixed integer program	An exact approach using special ordered sets and a heuristic using Lagrangian Relaxation
[31]	Mathematical modeling	Genetic algorithm
[32]	Behavioral experiments	Analytical analysis
[33]	Stochastic Models	Discrete-event simulations
[34]	Empirical research based on qualitative analysis	Empirical research based on qualitative analysis
[35]	Mathematical model based on Markov Chain	Analytical derivations
[36]	Integrated lot sizing and safety stock placement problem	Dynamic programming algorithm
[37]	A stochastic integer programming model	Modified stochastic genetic algorithm
[38]	The nonlinear model is formulated as a conic quadratic mixed-integer program	Solver is used to solve the optimization problem
[39]	A two-echelon supply chain model is considered under the newsvendor framework	Kuhn–Tucker methodology and classical optimization
[40]	A stochastic programming model	Solvers and valid inequalities
[41]	Dynamic programming	Dynamic programming algorithm
[42]	A mixed integer nonlinear programming	A piecewise linear approximation method and a simulated annealing heuristic approach
[43]	A mixed integer programming (MIP) formulation	A greedy heuristic algorithm
[44]	Mixed integer programming model	A two-stage solution process by solving two optimization models: a transportation model and a variation in the vehicle routing model

Table 2. *Cont.*

Work	Modeling Approach	Solution Method
[45]	A stochastic integer programming model	Approximation algorithm
[46]	Multi-location newsvendor problem with a game theoretical approach	A weighted proportional allocation rule and characterize the Nash equilibrium of the resultant ordering game among the store managers
[47]	Distributionally robust multilocation problem on a multilocation newsvendor network	A heuristic approximation and upper bounds
[48]	Behavioral model	Behavioral studies
[49]	Empirical research based on quantitatively analysis	Quantitatively describing the characteristics and tendencies of the samples
[50]	Service level optimization model	Approximation
[51]	A novel stochastic inventory optimization model.	Simulation
[52]	Optimization models and conic quadratic mixed-integer programs	Solved by using available solvers
[53]	Conduct a controlled between-subjects experiment with four treatments	Experimental hypotheses rely on the normative theory
[54]	Nonlinear optimization models	Analytical derivations and simulation
[55]	A two-stage stochastic program	Sample average approximation
[56]	Combining a mathematical planning model with a green allocation strategy	Genetic algorithm
[57]	A novel two-stage stochastic programming model	Sample average approximation
[58]	Systematic investigation	Systematic investigation and analysis
[59]	Mixed-integer linear programming and a greedy heuristic	Combination of mixed-integer linear programming and a greedy heuristic
[60]	A mixed-integer conic quadratic program transformed into a convex problem	Used standard optimization software
[61]	Newsvendor-based inventory model with partial pooling	Pragmatic heuristic inventory solution
[62]	A two-stage analytical metamodel	A metamodel and simulation-based optimization
[63]	Analytical models	Constraint generation and parameter search algorithm and a heuristic
[64]	Dynamic programming	Heuristic
[65]	A mixed-integer linear programming	A hybrid genetic algorithm
[66]	Empirical research based on qualitative analysis	Qualitative analysis
[67]	Combination of neural networks and mixed integerlinear programming model	Heuristic

3.5. Commonly Used Modeling Frameworks

In this section, we provide commonly used modeling frameworks in the reviewed studies. Table 3 summarizes these frameworks utilized in the related work. We classify the most frequently used three modeling frameworks as facility location, newsvendor framework, and lateral transshipments and label the studies that utilize them with “✓”. In the others category, we have game theory and central coordination. A significant amount of the examined papers includes central coordination for pooling activities.

Table 3. Commonly used modeling frameworks in reviewed articles.

Work	Lateral Transshipment	Facility Location	Newsvendor Framework	Others
[14]		✓		
[15]		✓		
[16]	✓			
[17]		✓		
[18]	✓	✓		
[19]				✓
[20]				✓
[21]	✓			
[22]				✓
[23]				✓
[24]				✓
[25]			✓	
[26]				✓
[27]				✓
[28]				✓
[29]				✓
[30]		✓		
[31]	✓			
[32]			✓	
[33]				✓
[34]		✓		
[35]				✓
[36]				✓
[37]				✓
[38]		✓		
[39]			✓	
[40]		✓		
[41]				✓
[42]		✓		
[43]	✓			
[44]				✓
[45]				✓
[46]			✓	
[47]			✓	
[48]			✓	
[49]	✓			
[50]				✓
[51]				✓
[52]				✓
[53]			✓	
[54]	✓			

Table 3. *Cont.*

Work	Lateral Transshipment	Facility Location	Newsvendor Framework	Others
[55]			✓	
[56]				✓
[57]			✓	
[58]				✓
[59]				✓
[60]				✓
[61]			✓	
[62]		✓	✓	
[63]		✓		
[64]		✓		
[65]		✓		
[66]				✓
[67]	✓			

Facility location is the first modeling framework used in the examined studies, and it is fundamental to effective inventory pooling strategies. These problems consider optimizing the locations of the facilities to be opened and their number in a supply chain network. In this way, the organizations can effectively manage the inventory levels, reduce the related operating costs, and improve their service levels. In this framework, central coordination on inventory levels could be achieved more easily. In the examined papers, 13 out of 54 articles consider facility location problems.

The newsvendor framework is commonly used in inventory management problems when demand is uncertain. In these problem settings, generally a single order is made for a perishable or seasonal product like newspapers, fashion goods, or foods, where the excess inventory cannot be sold. Inventory pooling applications in such a problem context is crucial since it helps to eliminate carrying the excess inventory which will be disposed of after a certain period. In the examined studies, 10 articles out of 54 consider a newsvendor framework.

Lateral transshipment corresponds to the transfer of the inventory between the locations on the same echelon, i.e., warehouses, retailers, or distribution centers. In this problem setting generally, the multiple locations share the inventory to optimize their service levels and reduce the related operating costs. The sharing mechanism among the participating entities can be different based on the problem context, such as complete and partial pooling. In complete pooling, the resources are shared among the participants without any restrictions, and the total inventory is considered as a central pool. On the other hand, in partial pooling, the participants can reserve some amount of the inventory themselves and partially contribute to the pool. In the examined studies, 8 out of 54 papers consider lateral transshipments.

While these modeling frameworks offer robust solutions, they are not without practical limitations. One major challenge is the significant computational effort required to solve models with high complexity, especially those incorporating stochastic elements or real-world variability. For instance, many studies employing stochastic models rely on approximations or heuristics to address computational infeasibility when exact solutions are impractical. Another key limitation is the reliance on extensive and high-quality data. Models often require detailed information on demand patterns, lead times, and cost param-

eters, which may not always be readily available or accurate in real-world scenarios. For example, studies that incorporate facility location models depend heavily on precise geographic and logistical data, while lateral transshipment frameworks require comprehensive data on inventory levels and inter-location dynamics.

3.6. Pooling Effects

In this section, after examining the reviewed studies, we provide an analysis on whether proposed methods of pooling are beneficial for the specific environments under consideration. Figure 7 presents a pie chart that demonstrates whether the proposed pooling strategies are beneficial, or beneficial under some conditions. Studies that do not specifically evaluate the performance of the pooling are classified as NA. These articles generally incorporate the pooling concept partially within the framework of a broader problem. Although the pooling strategy is generally classified as beneficial, most of the articles state that its benefits depend on some specific circumstances.

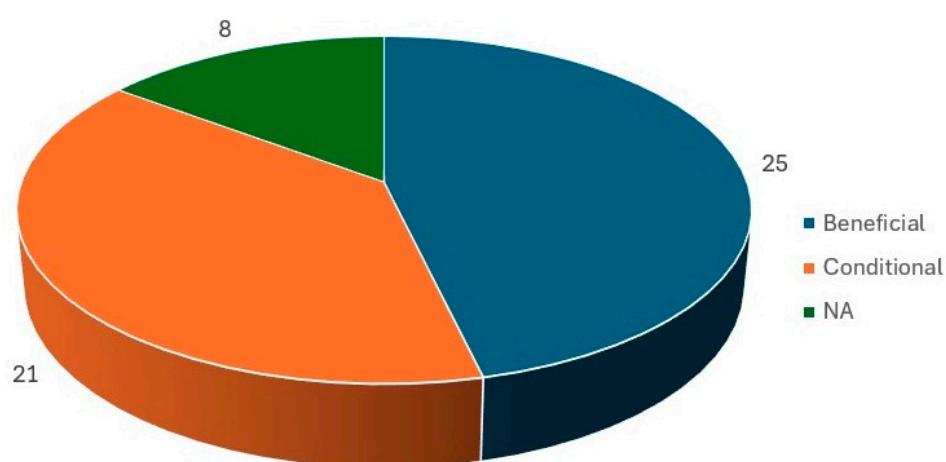


Figure 7. Pie chart indicating pooling concepts considered in the reviewed articles are beneficial or conditionally beneficial.

Pooling is generally found beneficial in terms of cost savings achieved through reduction in the safety stock levels, improved service levels, lead time reductions, and decreased shortage-related costs. On the other hand, most of the studies address that pooling may not always be beneficial if some conditions are not satisfied. Although there are various conditions across different problem contexts considered in the reviewed studies, we classify the most common and crucial ones to provide some insights. Most of these conditions are related to budget constraints, demand characteristics (variability, correlation, distributional characteristics like skewness, kurtosis, etc.), shared information quality, cost-related parameters, lead time durations, competition or cooperation among participating entities, incentives provided to participants, and sharing mechanisms.

3.7. Analysis on Sustainability

In this section, we analyze the selected articles based on their consideration of sustainability and related concepts. Sustainability refers to the principle of maintaining or enhancing processes over the long term without compromising the ability of future generations to meet their own needs. Inventory pooling, a critical inventory management strategy, involves consolidating the inventory across different locations or entities to achieve cost savings and improve service levels. This strategy not only focuses on optimizing operational efficiency but also aligns with core sustainability goals, such as reducing waste, enhancing resource utilization, and minimizing environmental impacts. Therefore, inventory pooling

is inherently linked to sustainability across its core dimensions: economic, environmental, and social sustainability.

Initially, we categorize the selected studies based on whether sustainability is addressed explicitly or implicitly. Six out of the fifty-four articles explicitly consider sustainability, while the contributions of the remaining articles are implicit. In this categorization, articles that explicitly consider sustainability highlight the direct contributions of their research, while the contributions of other articles require interpretation.

In the explicit category, four of the six articles consider returnable/reusable transport items (RTIs) pooling. Lakhmi et al. [59] consider RTI pooling for different customers and focus on the planning of empty items by optimizing the transportation routes. Liu et al. [44] also study the pooling of RTIs, and they consider inventory sharing, distribution, and routing in their model. Miao et al. [65] examine pallet pooling as a sustainable method for efficient utilization of the resources. They focus on locating the pallet pooling centers and optimizing the logistics operations. Yu et al. [56] investigate pallet pooling for a ship-building company. Instead of stocking, they propose reusing and sharing the items for sustainable practices.

The remaining two articles focus on different aspects of inventory pooling. The first article, by El Moussaoui et al. [66], examines a behavioral study on information sharing and its economic and environmental benefits within supply chains for the Morocco region. The second article, by Qi et al. [63], proposes a joint location–inventory model for pooling the charging of electrical batteries, aiming to enhance efficiency and sustainability in its domain.

Among these six articles, Miao et al. [65], Liu et al. [44], and Lakhmi et al. [59] warrant particular attention due to their specific contributions to sustainability. Miao et al. [65] consider the planning of pallet pooling centers integrated with inventory optimization and state that their proposed method results in a 3.85% reduction in inventory costs and a 0.96% reduction in transportation costs compared to a two-stage planning model. Liu et al. [44] examine the pooling of reusable transport items (RTIs) and provide a case study of an RTI pooling company, demonstrating a 28.1% reduction in transportation costs through the implementation of their approach. Similarly, Lakhmi et al. [59] focus on RTI pooling for diverse customer types and report a 30% reduction in the number of trucks used and a 20% reduction in traveled distances. Additionally, they highlight that jointly managing dedicated and pooled RTIs could yield a 9% reduction in transportation expenses.

El Moussaoui et al. [66], focusing on a behavioral study, emphasize that reliable information-sharing strategies can lead to notable reductions in costs and transportation expenses through pooling. The remaining two studies, Yu et al. [56] and Qi et al. [63], focus on green operations and solutions in their respective problem contexts. Yu et al. [56] propose a green pallet allocation strategy incorporating green transportation solutions and operations. They state that the proposed method provides average cost savings from 56% to 65%. Qi et al. [63] contribute to sustainability by addressing the planning and management of electric vehicle battery charging and swapping operations as a green initiative.

Overall, the metrics used to evaluate sustainability contributions in these studies primarily include cost savings, reductions in transportation expenses, and decreases in traveled distances, highlighting the alignment between operational efficiency and environmental benefits.

All six of these papers emphasize contributions to economic and environmental sustainability, with Qi et al. [63] also making a slight contribution to social sustainability through increased battery availability. In the remaining 48 articles, the reader must infer the contributions to sustainability. In the following sections, we further classify the articles based on their treatment of the three pillars of sustainability: economic, environmental,

and social. This classification is presented in Table 4. Articles that contribute to the corresponding pillars indirectly are labeled accordingly with “✓” for clearer representation.

Table 4. Classification of the articles based on their consideration of three pillars of sustainability: economic, social, and environmental.

Work	Economic	Social	Environmental
[14]	✓		✓ (indirectly)
[15]	✓		✓ (indirectly)
[16]	✓		✓ (indirectly)
[17]	✓		✓ (indirectly)
[18]	✓	✓	✓ (indirectly)
[19]	✓		
[20]	✓		✓ (indirectly)
[21]	✓	✓	✓ (indirectly)
[22]	✓		
[23]	✓		
[24]	✓		✓ (indirectly)
[25]	✓		
[26]	✓		
[27]	✓	✓	
[28]	✓		
[29]	✓		
[30]	✓	✓	
[31]	✓		
[32]	✓	✓ (indirectly)	
[33]	✓		✓ (indirectly)
[34]	✓		✓ (indirectly)
[35]	✓	✓ (indirectly)	✓
[36]	✓		
[37]	✓	✓	
[38]	✓		✓ (indirectly)
[39]	✓		
[40]	✓	✓ (indirectly)	
[41]	✓		
[42]	✓	✓	✓ (indirectly)
[43]	✓		
[44]	✓		✓
[45]	✓		
[46]	✓		
[47]	✓		✓ (indirectly)
[48]	✓		
[49]	✓	✓	
[50]	✓		
[51]	✓	✓	

Table 4. *Cont.*

Work	Economic	Social	Environmental
[52]	✓		✓ (indirectly)
[53]	✓		
[54]	✓	✓	
[55]	✓		
[56]	✓		✓
[57]	✓		✓ (indirectly)
[58]	✓	✓ (indirectly)	
[59]	✓		✓
[60]	✓		✓ (indirectly)
[61]	✓		
[62]	✓		✓ (indirectly)
[63]	✓	✓	✓
[64]	✓		✓
[65]	✓		✓
[66]	✓		✓
[67]	✓		✓ (indirectly)

3.7.1. Economic Sustainability

All the articles included in this review contribute to economic sustainability by considering inventory pooling, sharing and planning activities. Inventory pooling is a key method with the potential for cost savings, enhancing resource utilization, and improving service levels. As such, the articles we reviewed highlight the economic benefits of pooling and its role in sustaining the operations. In Section 3.6, we present a pie chart showing that 46 out of 54 articles classify the effects of pooling as either beneficial or conditionally beneficial. In the remaining eight articles, the effects of pooling are not explicitly investigated; however, they highlight the impact of the proposed methods on improving operational efficiencies, thereby contributing to economic sustainability.

Among the 54 articles, we analyzed those that highlight the benefits of pooling based on the metrics utilized in these studies. Several studies, including [17,22,28,31,34,42,45,56,67], emphasize cost savings achieved through pooling, with reported percentages ranging approximately from 10% to 60%. Savings resulting from reductions in inventory levels are discussed in studies such as [14,15,20,29,36,40,50], with [29] reporting a 20% reduction and [50] noting approximately 15%. Studies [44,59] specifically highlight cost savings through reductions in transportation costs, with reported savings of 28% and 9%, respectively. In contrast to cost-related metrics, refs. [21,50] emphasize improvements in service levels, with [21] also noting reduced wastage in the blood supply chain. Furthermore, refs. [39,53] focus on profit increases rather than cost reductions. The remaining studies, such as [25,27,43], examine other aspects of savings, including reductions in production and return levels, shortage levels, and downtimes, respectively. Therefore, the benefits of pooling encompass various dimensions, including cost efficiency, service level improvements, and operational optimizations.

3.7.2. Environmental Sustainability

In this section, we analyze the articles based on their direct or indirect contributions to environmental sustainability. The utilization of reusable/returnable items plays a significant role in promoting environmental sustainability by reducing waste, conserving

resources, and minimizing reliance on single-use products. Usage of these items contributes to the reducing environmental impact associated with manufacturing, packaging, and disposal. Therefore, studies that incorporate reusable/returnable items are categorized as making a direct contribution to environmental sustainability [35,44,56,59,63,65]. Similarly, articles that explore the potential use of renewable energy sources [64] or focus on reducing fuel consumption and subsequent emissions [66] are also considered to have a direct contribution to environmental sustainability.

Articles that indirectly contribute to environmental sustainability typically emphasize cost-minimization aspects of the models, rather than directly addressing environmental impacts. Strategies such as minimizing distance-dependent transportation costs [33,47,52,60], pooling logistics operations [34], optimizing location and routing decisions to reduce delivery times and transportation costs [24,42,57,62] and considering location–inventory [17,38], facility location [14,18] and dynamic allocation [67] models with minimizing distance-dependent transshipment costs help reduce unnecessary transportation and, consequently, contribute to lowering carbon emissions. In these studies, an indirect goal of the proposed models is associated with lowering carbon emissions. However, as highlighted by [14], pooling strategies involving reduced stock points may result in increased transportation requirements compared to the no-pooling scenarios. This potential downside underscores the importance of carefully evaluating the trade-offs between centralized stock management and the associated transportation impacts. While the primary aim of these studies is to develop cost-effective pooling strategies, increased transportation needs stemming from reduced stock points offset environmental gains. To address these concerns, future research should explore mitigation strategies such as eco-friendly transportation options and advanced routing optimization techniques to balance these trade-offs and achieve both cost and environmental sustainability goals. The remaining indirect contribution to environmental sustainability is the reduction in resource wastage and shortages [15,16,20,21] through the minimization of transportation costs.

3.7.3. Social Sustainability

Social sustainability is focused on ensuring equity, justice, and well-being for all, promoting fair access to resources and opportunities for both present and future generations. The articles that directly contribute to social sustainability focus on reducing shortages and increasing the availability of public health services [21,27,37,42,49,51,54], as well as improving supply chain resilience to ensure service continuity [18,30,63]. The remaining articles that indirectly contribute to social sustainability address issues such as enhancing item availability for pooling participants [35], ensuring reliable information sharing among supply chain entities [32], increasing disaster resilience to facilitate efficient allocation of relief supplies [40], and investigating the equity within the supply chain members [58].

3.8. Challenges in Practice

Although inventory pooling strategy is generally regarded as beneficial in many problem contexts, the implementation phase creates numerous challenges for the practitioners. In this section, we highlight several of these difficulties and provide some remedies presented in the reviewed studies. The first of them is related to the data and the shared information quality. In an inventory pooling environment with multiple participating entities, the primary goal of the participants is to maximize their own profits. Therefore, in these kinds of environments, participants may not be willing to exchange information or distort some of their private data in order to gain more profit from the coalition. These challenges are addressed through behavioral studies [32,48,49,53].

A remedy for data and shared information quality is to utilize radio frequency identification (RFID) technology for real-time data tracking and sharing. In this way, the shared data will be available for all participants in the coalition at any time. This technology not only contributes to building trust among participants but also facilitates pooling operations by improving data quality. Using radio frequency identification (RFID) technology to achieve real-time data of the items is considered in [25,39]. Another remedy to increase the willingness of the information sharing is incentivizing the participants to share their private information so that being in the coalition becomes more profitable.

Another challenge faced by the practitioners is the complexity of the constructed models for the problem environment. The efficient implementation of the inventory pooling strategy requires creating a model that is responsive to the changing dynamics of the supply chain in consideration. Therefore, the dynamic nature of real-life practical problems leads to complex models that may not always be solved efficiently to optimality. Therefore, by examining Table 2, it is clearly seen that most of the solution methods utilized various approximations, heuristics, and simulations. These are the common solution approaches used when obtaining exact solutions to the proposed models is either time-consuming or impractical. Instead of finding optimal solutions, these methods provide good approximate solutions in considerably less time. In this way, the large instances of real-life problems could be solved effectively.

4. Future Research Areas

In this section, we present potential areas for future research based on an examination of real-life application papers from the past 14 years. Given the rapid advancements in artificial intelligence and big data, future studies that incorporate these techniques into real-world applications will be essential. Specifically, methods such as reinforcement learning (RL) and stochastic optimization can offer dynamic solutions for inventory pooling under uncertain demand conditions. For instance, RL algorithms could optimize inventory allocation and routing in real time, adapting to complex and fluctuating supply chain environments. These techniques may also provide significant advantages in modeling non-stationary or highly variable demand patterns.

Another promising approach is the development of distribution-free learning methods, which do not rely on strict probabilistic assumptions. These methods hold significant potential for real-life problems where demand is uncertain or data are incomplete. By relaxing assumptions on demand distributions, distribution-free methods can improve decision-making processes, making them applicable to a broad range of complex industry problems. Research questions for future exploration could include: How can distribution-free learning methods outperform traditional approaches in managing inventory pooling under limited data availability?

A further dimension for future research lies in creating new and fast computational techniques to address the increasing complexity of real-world environments. Efficient implementation of inventory pooling strategy requires responsive modeling approaches capable of handling rapidly changing characteristics of the supply chain dynamics. Integrating deep learning algorithms with traditional optimization techniques, such as mixed-integer programming or metaheuristics, could be a promising direction. For example, hybrid models could reduce computational effort while maintaining high accuracy, enabling large-scale pooling problems to be solved efficiently. Research could focus on questions like: What role do hybrid AI-based models play in enhancing computational efficiency for large-scale inventory pooling scenarios?

A critical area for future research is the explicit integration of sustainability goals into inventory pooling models. In our sustainability analysis, only a few articles—six out of fifty-

four—explicitly consider sustainability in their research. The remaining 48 articles focus more on cost-effectiveness rather than on sustainability. Sustainability is a critical consideration in supply chain management and logistics that can balance economic performance with environmental and social impacts. Future studies could leverage multi-objective optimization techniques to address trade-offs between economic, environmental, and social sustainability. Research questions might include: How can inventory pooling strategies minimize carbon footprints while maintaining cost efficiency? or What mechanisms can incentivize stakeholders to adopt pooling models that prioritize long-term sustainability?

Finally, sector-specific applications provide another avenue for meaningful research. Industries such as healthcare, retail, and energy could benefit from tailored pooling strategies that address unique challenges. For example, future research could explore: How can AI-driven inventory pooling improve healthcare supply chain resilience during pandemics? or What pooling mechanisms can optimize renewable energy resource sharing across distributed grids?

By addressing these targeted areas, future research can not only enhance the theoretical understanding of inventory pooling but also provide actionable solutions to improve its practical implementation across diverse and dynamic industries.

5. Conclusions

In this review paper, we have examined inventory pooling practices in real-life applications by analyzing the articles published over the past 14 years across various sectors, such as healthcare, retail, transportation, and logistics. We classified the relevant literature based on sectors, product/service types, data characteristics, modeling and solution approaches, and finally, contributions to sustainability. Our categorization emphasizes the breadth of inventory pooling implementations. The findings suggest that inventory pooling has a prominent potential for reducing costs, enhancing service levels, and increasing the efficiency of supply chains. However, its effectiveness depends on certain characteristics of the problem context, such as demand patterns, product types, and supply chain dynamics.

Over the past 14 years, no single modeling approach stands out, and this reflects the complexity and diverse nature of inventory pooling problems. On the other hand, stochastic models are predominantly used due to inherent demand uncertainties encountered in real-life problems. Heuristics and approximation-based solution techniques are frequently utilized to address computational difficulties, especially when obtaining optimal solutions is time-consuming. This trend aligns with the growing complexity of inventory pooling in real-life applications, where obtaining exact solutions is often impractical.

While inventory pooling has demonstrated its ability to optimize operational efficiency and cost-effectiveness, the integration of sustainability goals remains limited. Only a small subset of the reviewed studies explicitly incorporates environmental or social sustainability considerations. As supply chains face increasing pressure to address environmental and social challenges, future research must prioritize embedding sustainability directly into inventory pooling strategies. This can be achieved through multi-objective optimization models that balance economic performance with reductions in carbon emissions, resource conservation, and social equity.

The integration of recent emerging technologies, such as artificial intelligence and learning algorithms, into real-life inventory pooling problems offers significant potential as a future research area. Utilizing these methods in real-life environments could provide greater flexibility in decision-making processes, thereby enabling more effective implementations of inventory pooling. Leveraging deep learning algorithms to achieve faster and good approximate solutions offers another promising avenue for future research. While inventory pooling has demonstrated significant potential for effective implementations

over the past 14 years, further exploration of its application alongside these advanced technologies will be critical to fully unlock its potential in real-life scenarios. Moreover, incorporating sustainable practices into these advanced models is essential, and studies focusing on real-life applications that integrate sustainability with inventory pooling are needed to ensure long-term environmental, social, and economic benefits.

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